

LED Flasher PCB Design

Eclectronics Fall 2026 at Olin College

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1 Overview

The goal of this assignment is to design a small LED-flasher PCB as the kickoff project for the Eclectronics course and to get up to speed on the board-design tools (KiCad and LTspice) used for the rest of the semester. The board runs off a USB type-A port and flashes an LED at 1 Hz, within $\pm 10\%$.

The allowed parts list contains no clock or timer IC, so the oscillation has to come from resistors, capacitors, and an op-amp. The result is a hysteretic (Schmitt-trigger) relaxation oscillator. Figure 1 shows the finished board layout; the schematic is in fig. 2.

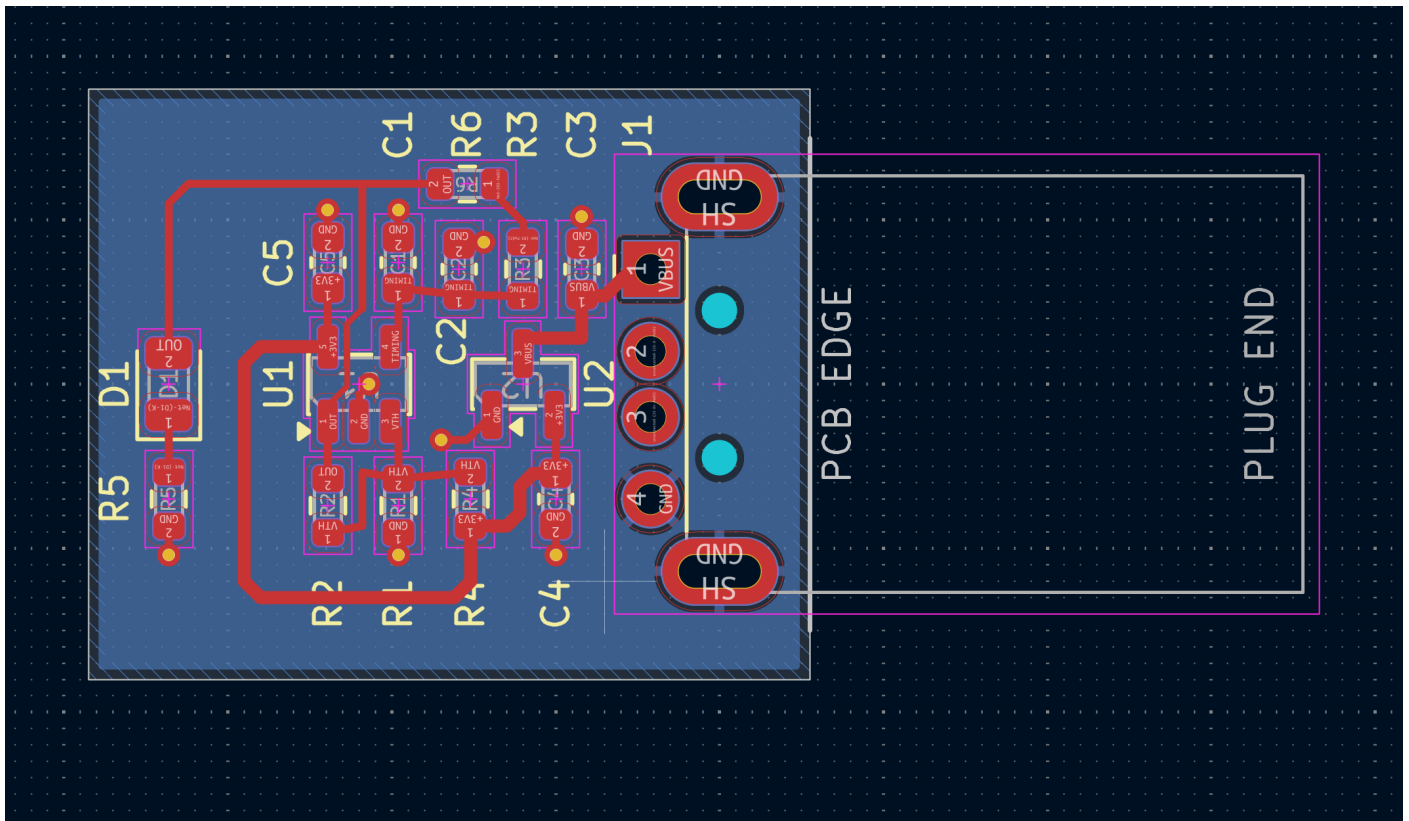


Figure 1: PCB layout in the KiCad editor, with the USB plug end to the right.

1.1 Requirements summary

1. **Flash period of $1\text{ s} \pm 10\%$.** The nominal design is 1.002 s. Section 3 shows the worst-case period over component tolerances is 0.929–1.078 s, within the window.
2. **Single-ended 3.3 V supply from USB 5 V VBUS.** An MCP1702-3302 linear regulator steps VBUS down to 3.3 V.
3. **Components only from the provided spreadsheet.** Every part in the BoM (section 5) is from the course list.
4. **Adequate bypass and bulk capacitance.** Bulk capacitors on the regulator keep it stable and hold the rail steady; a bypass capacitor at the op-amp supply pin keeps switching noise off the rail so it cannot jitter the oscillator thresholds.
5. **Two-layer PCB, all components on top, 6-mil trace/space, 24-mil vias with 12-mil drills, no vias in pads.** These constraints keep the board simple and cheap to build.

2 Circuit Design

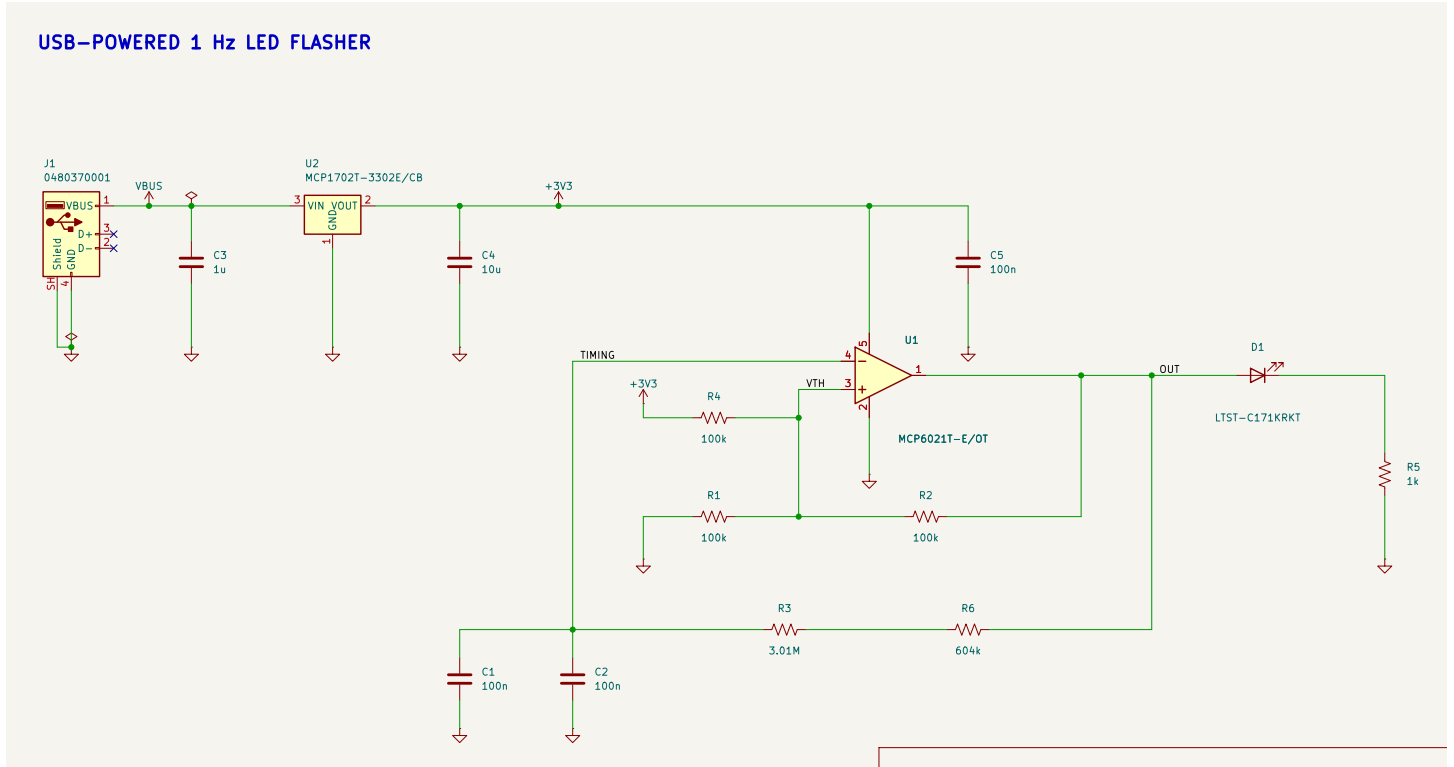


Figure 2: Full circuit schematic (KiCad).

2.1 Oscillator

The oscillator can be thought of as built up in three steps:

1. **Comparator.** An op-amp with no feedback is a comparator: its huge open-loop gain drives the output to one rail or the other depending on which input is higher. With a rail-to-rail part on a 3.3V supply the output is either ≈ 0 or $\approx V_{DD}$. The problem is that there is only one threshold, so a signal near it makes the output chatter on any noise.
2. **Schmitt trigger.** Adding positive feedback (a resistor from the output to the non-inverting input) makes the threshold depend on the current output state. When the output is high the threshold is V_{TH} ; when it is low the threshold drops to V_{TL} . The circuit remembers its last state, and the gap between the two thresholds is the hysteresis.
3. **Relaxation oscillator.** Adding negative feedback through a resistor R to a capacitor C on the inverting input turns the Schmitt trigger into an oscillator. When the output is high, C charges through R toward V_{DD} . Once it crosses V_{TH} the output flips low, the current reverses, and C discharges toward ground. When it falls below V_{TL} the output flips high again. The result is a square wave, which drives the LED.

Period and duty cycle. Each half of the cycle is the capacitor charging or discharging through R between the two thresholds, so

$$t_{\text{high}} = RC \ln \left(\frac{V_{DD} - V_{TL}}{V_{DD} - V_{TH}} \right), \quad t_{\text{low}} = RC \ln \left(\frac{V_{TH}}{V_{TL}} \right), \quad T = t_{\text{high}} + t_{\text{low}}. \quad (1)$$

R and C appear the same way in both halves, so they set how long the period is but not the duty cycle. The duty cycle comes from where the thresholds sit: if they are evenly spaced around the middle of the supply, the two halves take the same time.

In this design the three threshold resistors are all 100 k Ω , so the non-inverting input sits at the average of V_{DD} , ground,

and the output. That gives

$$V_{TH} = \frac{2}{3}V_{DD} = 2.20 \text{ V}, \quad V_{TL} = \frac{1}{3}V_{DD} = 1.10 \text{ V}, \quad (2)$$

which are evenly spaced around $V_{DD}/2$, so the duty cycle is 50 % and both logarithms in eq. (1) reduce to $\ln 2$:

$$T = 2RC \ln 2. \quad (3)$$

V_{DD} drops out, so the period does not depend on the exact supply voltage.

2.2 Power supply

The USB connector supplies 5 V on VBUS, and the circuit needs 3.3 V, so an MCP1702-3302 linear regulator steps it down. A linear regulator is appropriate because the load is only a few milliamps, so the 1.7 V drop dissipates negligible power, and the output is clean with no switching noise.

Bulk capacitors are placed on the regulator input ($C_3 = 1 \mu\text{F}$) and output ($C_4 = 10 \mu\text{F}$). The output capacitor is required by the MCP1702 datasheet for loop stability (minimum $1 \mu\text{F}$). Both also hold the rails steady when the LED switches, since the USB cable cannot deliver a current step instantly. This matters because the oscillator thresholds are derived from V_{DD} , so any sag would show up as period jitter. The op-amp gets its own 100 nF bypass (C_5) close to its supply pin.

2.3 Component selection

From eq. (3), a 1 s period needs $RC = 1/(2 \ln 2) = 0.721 \text{ s}$. The parts list limits how that product can be split up. The 100 nF capacitor was picked for timing because it has the tightest tolerance (5 %) of the larger values available; two in parallel give $C = 200 \text{ nF}$. That calls for $R \approx 3.61 \text{ M}\Omega$, and since no single resistor on the list is close, two are placed in series: $3.01 \text{ M}\Omega + 604 \text{ k}\Omega = 3.614 \text{ M}\Omega$. The nominal period is then

$$T = 2(3.614 \text{ M}\Omega)(200 \text{ nF}) \ln 2 = 1.002 \text{ s}. \quad (4)$$

The three threshold resistors are all 100 k Ω . Only their ratio matters for the thresholds, so a mid-range value was chosen that keeps the current through the divider small.

Component	Purpose
J1, USB connector	Supplies approximately 5 V and ground. The USB data pins are unused.
U2, voltage regulator	Converts USB power into the 3.3 V supply for the oscillator.
C3, 1 μF	Input bypass/decoupling capacitor. Provides local charge at the regulator's input and helps reduce supply disturbances.
C4, 10 μF	Bulk capacitor and regulator output capacitor. Stores charge to support changes in current demand and helps keep the 3.3 V regulator output stable.
C5, 100 nF	Local bypass/decoupling capacitor for U1. Handles fast supply disturbances and brief current demands. It belongs physically close to U1's power pin.
U1, op-amp	Compares the timing voltage at its $-$ input with the threshold at its $+$ input. Switches its output high or low to drive the LED and the timing network.
R1, 100 k Ω	Pulls the threshold voltage toward ground. Works with R2 and R4 to establish the switching thresholds.
R4, 100 k Ω	Pulls the threshold voltage toward 3.3 V. Works with R1 and R2 to establish the switching thresholds.
R2, 100 k Ω	Provides positive feedback, creating two switching thresholds, 1.10 V and 2.20 V. This is the circuit's hysteresis.
R3, 3.01 M Ω and R6, 604 k Ω	Together form 3.614 M Ω of timing resistance. Control how quickly C1/C2 charge and discharge.
C1 and C2, 100 nF each	Timing capacitors, not bypass or bulk capacitors. Their parallel connection gives 200 nF. Their changing voltage determines the blink rate together with R3/R6.
D1, red LED	Lights when U1's output is high.
R5, 1 k Ω	Limits LED current and the load on U1's output.

Table 1: Purpose of each component.

2.4 LED drive

The LED is driven directly from the op-amp output through $R_5 = 1\text{ k}\Omega$. A red LED (LTST-C171KRKT) is used because it has the lowest forward voltage of the available colors, about 2.0 V, giving

$$I_{\text{LED}} = \frac{V_{\text{DD}} - V_F}{R_5} \approx \frac{3.3\text{ V} - 2.0\text{ V}}{1\text{ k}\Omega} \approx 1.3\text{ mA}. \quad (5)$$

This is bright enough for an indicator and well below the op-amp's output current limit, so the output still swings to the rail.

3 Sensitivity Analysis

The period has to land between 0.9 s and 1.1 s even when every part is at the edge of its tolerance. From eq. (3), $T = 2RC \ln 2$, so the period is directly proportional to R and to C : if R is 1 % high, T is 1 % high, and the same for C . The threshold resistors matter too, because they set how far the capacitor has to swing each cycle, but their effect is smaller. Writing the period as $T = RC \cdot k$, where k collects the threshold terms,

$$\frac{\Delta T}{T} = \frac{\Delta R}{R} + \frac{\Delta C}{C} + \frac{\Delta k}{k}. \quad (6)$$

Timing resistor and capacitor. R is two 1 % resistors in series and C is two 5 % capacitors in parallel, so at worst R is off by 1 % and C by 5 %. Together:

$$\begin{aligned} \text{low: } & (0.99)(0.95) = 0.9405 \Rightarrow -5.95\%, \\ \text{high: } & (1.01)(1.05) = 1.0605 \Rightarrow +6.05\%. \end{aligned}$$

Threshold resistors. With all three at exactly 100 k Ω , $k = 2 \ln 2 = 1.3863$. Putting each one at $\pm 1\%$ in the worst combination moves k to 1.4064 (+1.45 %) or 1.3664 (−1.44 %).

Worst case. Stacking both effects in the same direction gives 0.929–1.078 s (table 2), about −7.3 % / +7.6 %. This is inside the $\pm 10\%$ requirement with roughly 2.5 % to spare. The capacitor tolerance is by far the largest contributor.

Case	RC factor	k	T (s)	Error (%)
Nominal	1.0000	1.3863	1.0020	0.00
RC low only	0.9405	1.3863	0.9424	−5.95
RC high only	1.0605	1.3863	1.0626	+6.05
All low (worst-case minimum)	0.9405	1.3664	0.9289	−7.30
All high (worst-case maximum)	1.0605	1.4064	1.0780	+7.59
Specification limits			0.9 / 1.1	± 10

Table 2: Period at tolerance corners. Nominal $RC = 0.7228\text{ s}$.

4 LTspice Simulation

The oscillator was also simulated in LTspice, mainly for practice with the tool. The simulation (fig. 3) is the same circuit as the board with a generic op-amp model and an ideal 3.3 V source in place of the USB input and regulator. It was run before the final timing parts were chosen, so it uses $R = 698\text{ k}\Omega$ and $C = 1\text{ }\mu\text{F}$; from eq. (3) the predicted period for those values is $2(0.698) \ln 2 = 0.968\text{ s}$.

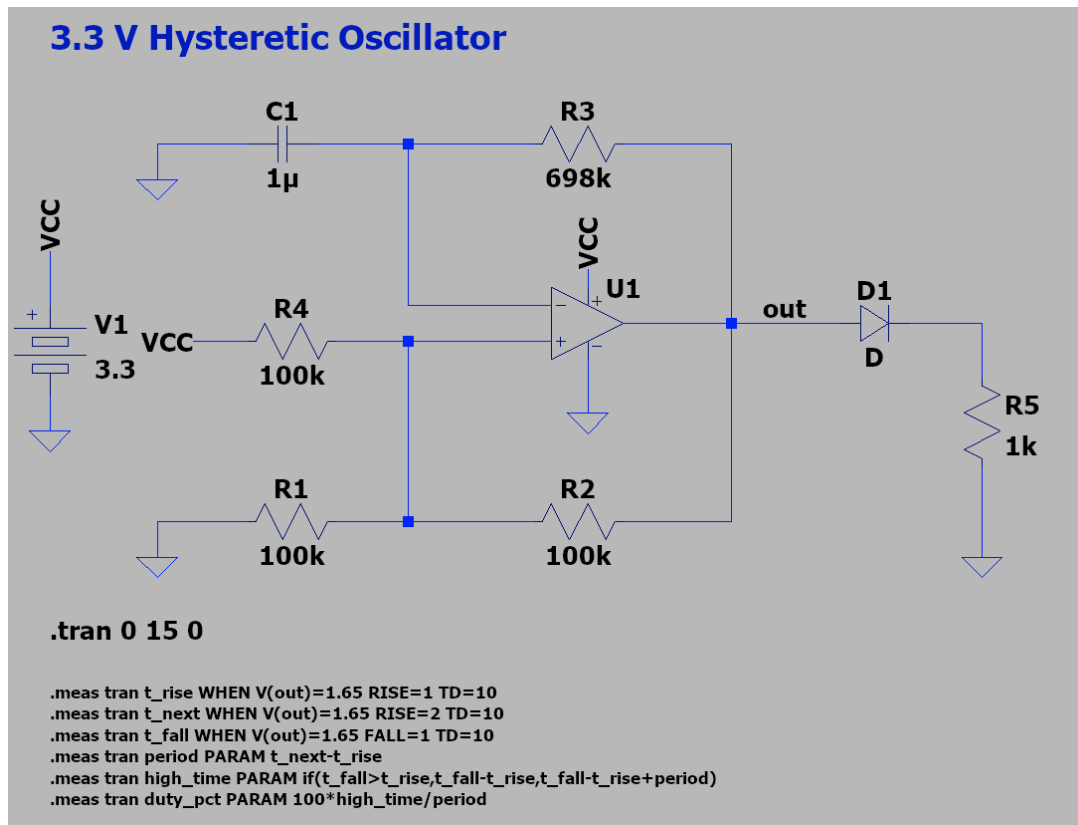


Figure 3: LTspice schematic with .meas statements for period and duty cycle.

Figure 4 shows the output. The simulation reports a period of 0.963 s and a duty cycle of 50.2 %, within 0.5 % of the prediction. The output sits at 1.65 V for the first few seconds because the simulation starts with both op-amp inputs at exactly the same voltage and there is no noise to tip it one way; a real circuit starts immediately.

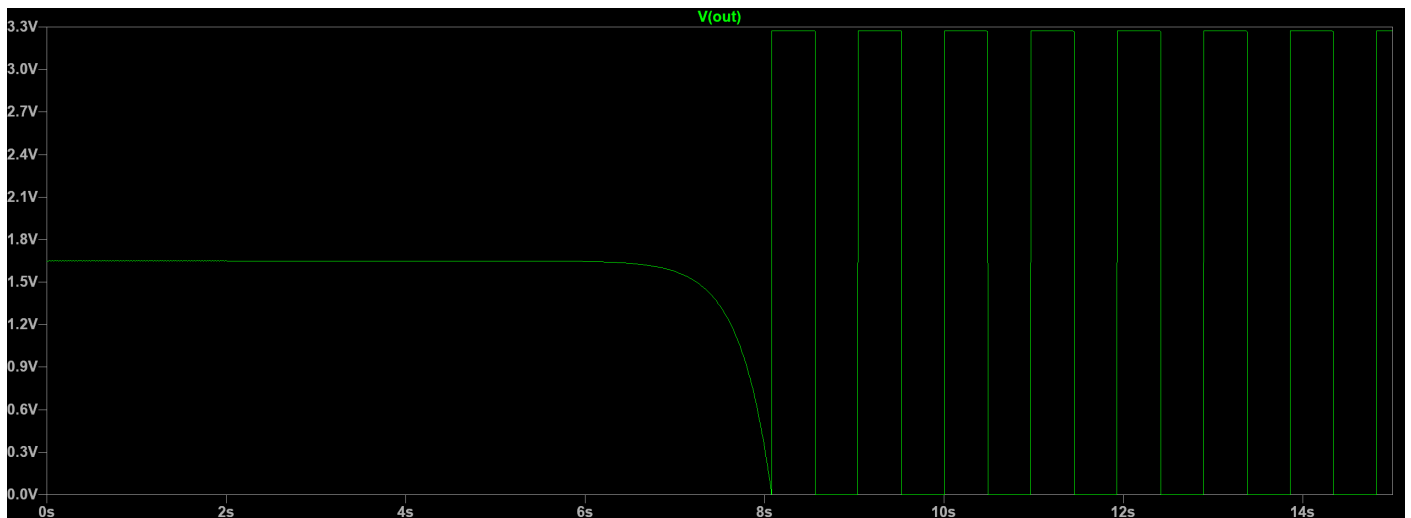


Figure 4: Simulated output voltage.

5 Bill of Materials

All parts are drawn from the course spreadsheet. Reference designators match the schematic in fig. 2.

Ref. des.	Description	Qty	Package	Mfr. part #	Digi-Key #
J1	USB-A plug, right angle, through-hole	1	–	0480370001	WM17117-ND
U1	Op-amp, 10 MHz RRO, single	1	SOT-23-5	MCP6021T-E/OT	MCP6021T-E/OTCT-ND
U2	LDO, 3.3 V, 250 mA	1	SOT-23-3	MCP1702T-3302E/CB	MCP1702T-3302E/CBCT-ND
D1	LED, red, clear	1	0805	LTST-C171KRKT	160-1427-1-ND
C1, C2, C5	100 nF X7R 50 V 5 %	3	0603	CC0603JRX7R9BB104	311-1779-1-ND
C3	1 μ F X7R 25 V 10 %	1	0603	C0603C105K3RACTU	399-7376-1-ND
C4	10 μ F X5R 16 V 10 %	1	0603	GRT188R61C106KE13D	490-12317-1-ND
R1, R2, R4	100 k Ω 1 %	3	0603	RC0603FR-07100KL	311-100KHRCT-ND
R3	3.01 M Ω 1 %	1	0603	RC0603FR-073M01L	311-3.01MHRCT-ND
R5	1 k Ω 1 %	1	0603	RC0603FR-071KL	311-1.00KHRCT-ND
R6	604 k Ω 1 %	1	0603	RC0603FR-07604KL	311-604KHRCT-ND

Table 3: Bill of materials (15 components, 11 line items).

6 PCB Layout

All components are on the top layer. Signals are routed on top and the bottom layer is a solid ground pour, with a via at each ground pad. The USB connector sits on the bottom edge with the regulator and its capacitors right behind it, the op-amp is in the center with its bypass capacitor beside its supply pin, and the timing parts are grouped together next to it. The LED is at the top edge so it is visible when plugged in. Table 4 summarizes the layout against the design rules; DRC passes with no violations.

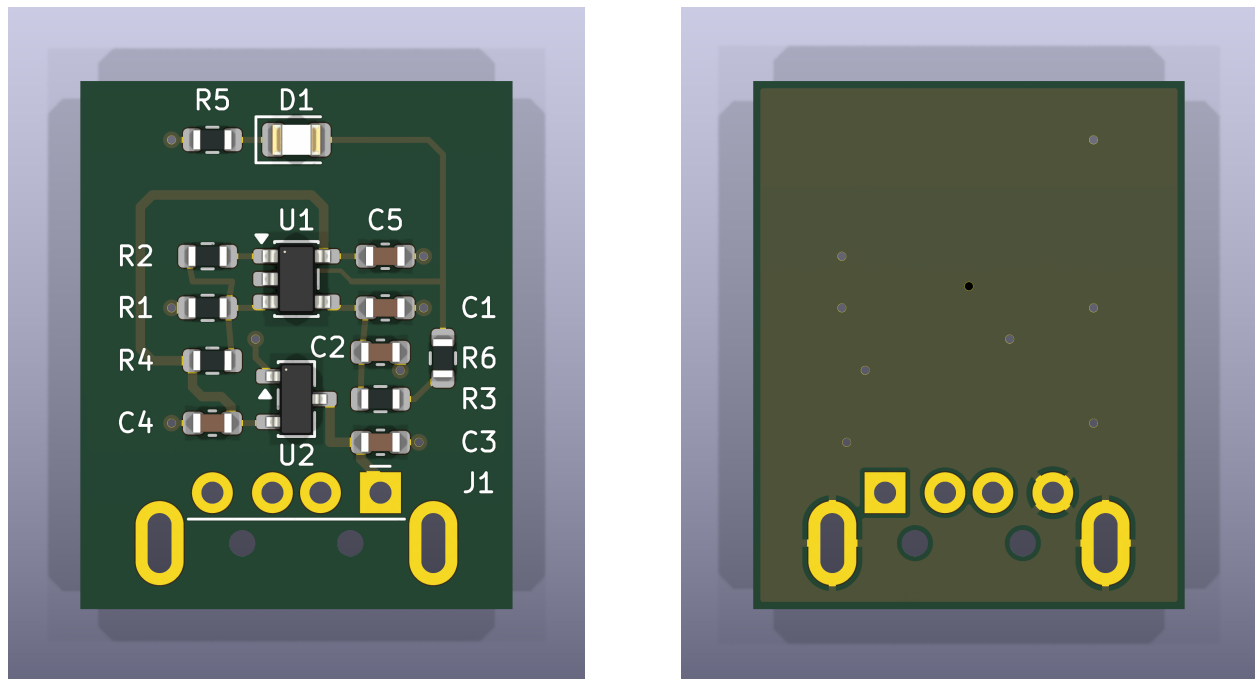


Figure 5: 3D render of the board: top side with components (left) and bottom ground pour (right).

Metric	Value
Board width (mm)	18
Board height (mm)	22
Bounding-box area (mm ²)	396
Min. trace width used (mil)	7.09
Via count	9
DRC violations	0

Table 4: Layout summary.

7 Design Files

All design files, including the KiCad project, LTspice simulation, and this report, are in the GitHub repository at <https://github.com/flipper-slipper/eclectronics-fall-2026>.